Data Analysis Programming
Semester 2026-II
Workshop No. 1 — Hazard Definition, Source Validation, and Data Contract

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Welcome to the first workshop of the *Data Analysis Programming* course!

This workshop starts the semester-long project: a multi-source ingestion and analysis system for one **natural hazard**. Before writing any pipeline, each team must prove that its data sources work, understand what they contain, and decide how everything will fit in a single canonical data model. The decisions made here are the foundation of Workshops #2–#4, so changes at this stage are far cheaper than later.

General Workshop Definition: Source research and data contracting involve identifying, testing, and documenting the data sources that will feed the system, mapping each of them to a common schema, and formalizing the project (users, goals, scope, and risks). The result is a verified plan, not a promise: every source in the plan must have been queried for real.

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Workshop Scope and Objectives

- **Hazard Understanding:** Explain the physical phenomenon, how it is measured, the meaning and units of its severity measure, and why monitoring it matters.
- **Source Identification and Validation:** Select and test **two historical** sources (different formats or APIs) and **two periodically updated** sources.
- **Source Profiling:** Characterize volume, time coverage, update frequency, identifiers, coordinates, units, time zones, missing values, and licensing.
- **Canonical Mapping:** Define how each source maps onto the common `event` model and which domain tables are needed.
- **Ingestion Contract Draft:** Define, per source, the historical strategy and the incremental strategy (what is the watermark?).
- **Project Formalization:** Functional user, user stories, indicator proposal, scope, and limitations.

Source Research and Characterization Methodology

1. **Hazard Definition:**
   - Describe the phenomenon and its severity measure (e.g., magnitude, rainfall accumulation, fire radiative power, wind speed, alert level) with units.
   - Define the geographic scope (global, a country, or a region) and the historical time window (e.g., 2010–present). Justify the choices in terms of data volume and relevance. The scope must be covered by *all four* sources: several sources are national or regional services (see *Geographic Scope and Source Coverage* in the Team Project Assignment).
   - State what *current situation* means for this hazard (last 24 hours, last 7 days, active alerts, etc.).
2. **Historical Source Research (two sources):**
   - Select two sources with **different formats or access methods** (for example, a REST/GeoJSON API and a bulk CSV archive).
   - Document for each: owner, URL, access method, authentication, rate limits, license and attribution requirements, format, fields, time coverage, and update policy.
   - Query each source for real and keep the unmodified response as evidence.
   - Estimate the full historical volume (rows and size) and decide how the bulk load will be partitioned (by year, month, or region) given API limits.

3. **Incremental Source Research (two sources):**
   - Select two sources that are updated continuously or periodically.
   - Characterize the update behavior empirically: poll twice, at least one hour apart (or on two different days for slow sources), and record what changed.
   - Identify how new or modified records can be detected: stable identifiers, "updated since" parameters, timestamps, ETag/Last-Modified headers, or content hashes.
   - Determine whether records can be *revised or deleted* after publication (for example, magnitudes that are updated or events that are removed) and what that implies for the watermark design.
4. **Source Profiling and Sample Validation:**
   - For each of the four sources, retrieve at least **200 records** (or the full source if smaller) and profile them in a notebook: null percentages per field, value ranges, duplicated identifiers, timestamp format and time zone, coordinate validity, and unit consistency.
   - Identify at least five concrete data-quality problems across the sources (and give an example record for each).
   - Verify *overlap*: do the sources describe some of the same events? Show at least five candidate pairs of records that you believe are the same real-world event and explain what differs (time, position, severity, identifier).
5. **Canonical Mapping:**
   - Produce a mapping table from each source field to the canonical `event` fields (`event_id`, `source`, `event_type`, `timestamp`, `latitude`, `longitude`, `severity`, `raw_reference`).
   - Define the hazard-specific attributes and propose the domain table(s) (e.g., `earthquake`: depth, magnitude type, place; `fire_detection`: brightness, confidence, satellite).
   - Define unit conversions and the rules for timestamps (UTC), coordinates (WGS84, decimal degrees), and severity scales.
6. **Ingestion Contract Draft:**
   - For each source write a short contract: historical strategy (partitioning, pagination), incremental strategy (watermark field, polling frequency, overlap window to catch late updates), and duplicate key.
   - Explain how the system will remain correct if a run fails halfway.

Project Definition Document

1. **Project Overview:** Title, motivation, goal (from a data-engineering and analytical perspective), scope, explicit exclusions, and constraints.

2. **Functional User:** Role (e.g., emergency-management analyst, risk researcher, journalist, local authority), profile, decisions supported, and why the dashboard is useful.
3. **User Stories:** At least six in the form *"As a [user], I want to [action] so that [benefit]"*, prioritized (MoSCoW). Cover monitoring (what is new), analysis (is this unusual?), and trust (how fresh and how complete is the data?).
4. **Indicator Proposal:** A preliminary definition of the hazard indicator (inputs, rule, output scale) and how it will be validated against historical events.
5. **Risks and Mitigation:** At least five (source outage, rate limits, schema change, license, data volume, ambiguous identifiers) with mitigations.
6. **Architecture Overview:** High-level diagram following the reference architecture in the Course Project document, adapted to the hazard.

Deliverables and Documentation

1. **Project Definition Report (PDF)** with all sections above, labeled diagrams, and real response snippets as evidence.
2. **Source Profile Notebook** (`notebooks/01_source_profiling.ipynb`) executed top to bottom with outputs, and the raw samples under `workshop_1/samples/` (small files; document how to download larger ones).
3. **Canonical Mapping Table** (`workshop_1/mapping.md` or CSV) and a first draft ER diagram.
4. **Repository Setup** with this structure:
   - `src/` — ingestion package (sources, normalization, storage)
   - `ingest.py` — command-line entry point (stub at this stage)
   - `config/` — `sources.yaml` with one entry per source (URL, format, schedule, watermark field)
   - `data/raw/` — raw landing zone (git-ignored except small samples)
   - `sql/` — schema definitions
   - `dashboard/` — Dash application
   - `tests/` and `tests/fixtures/` — tests and recorded sample responses
   - `notebooks/`, `docs/`, `workshop_1/`
   - `.env.example`, `.gitignore`, and a `README.md` with the hazard, team, sources, and structure

Submission Requirements

- Submit the report as a single PDF through the designated course platform, with the GitHub repository link.
- All documentation must be in **English** and follow professional technical writing standards.
- Cite every source and respect its license and attribution requirements.
- Do not commit secrets. API keys belong in environment variables.

Evaluation (20 points)

Criterion	Points
Hazard understanding and scope	2
Four validated sources with evidence	5
Source profiling and data-quality findings	4
Canonical mapping and domain tables	4
Ingestion contract (historical and incremental)	3
Project definition, user stories, and repository	2

Important Guidelines and Considerations

- **Validate Before Committing:** A source that looks ideal on paper but fails in practice blocks every later workshop. Real samples are your proof of concept.
- **Free and Open Sources Only:** Use sources freely accessible without payment. Check the license.
- **Be Polite to Servers:** Respect rate limits and terms of use; cache what you download and never re-download the historical archive needlessly.
- **Hazard Assigned, Sources Open:** The hazard is assigned by the professor; the suggested sources are a starting point. A different source is acceptable if it satisfies the requirements and is approved.
- **Do Not Over-Claim:** Monitoring and correlation are the goal. Do not present indicators as predictions of the hazard.
- **Request Keys on Day One:** Some sources need registration (NASA FIRMS `MAP_KEY`) or pre-approval (ReliefWeb `appname`). Request them immediately so that access delays do not block the workshop.
- **Think About Change:** Sources revise and withdraw records. Identify this early, as it determines the incremental design.

Start by running the queries yourselves: reading a real response teaches more than any documentation. Good luck!